

BRD (Business Requirement Document)

ELKash ERP System

1. Document Information

Item	Description
Project Name	ELKash ERP
Document Type	Business Requirement Document (BRD)
System Type	Modular ERP + POS
Target Industry	Cafe, Coffee Shop, Small Restaurant
Architecture	Web-Based Modular ERP
Frontend Stack	Nuxt 3, Vue 3, TypeScript
Backend Stack	Laravel REST API
Database	PostgreSQL (Supabase)
Authentication	Laravel Sanctum
Deployment	Netlify (Frontend) + VPS Ubuntu Nginx (Backend)
Version	V1
Status	Initial Draft

2. Business Background

Banyak bisnis cafe dan restoran skala kecil hingga menengah masih menggunakan proses operasional yang terpisah antara pencatatan transaksi, stok barang, pelaporan penjualan, dan pengelolaan user.

Sebagian besar bisnis menggunakan kombinasi:

- aplikasi kasir sederhana,
- spreadsheet manual,
- pencatatan stok non real-time,
- dan pelaporan yang tidak terintegrasi.

Kondisi tersebut menyebabkan:

- inkonsistensi data,
- human error,
- keterlambatan laporan,
- sulitnya monitoring bisnis,
- dan minimnya kontrol operasional.

ELKash dikembangkan sebagai solusi ERP modular modern yang mengintegrasikan:

- POS,
- inventory,
- reporting,
- user management,
- dan business workflow dalam satu sistem terpusat.

Sistem dirancang dengan pendekatan realistic enterprise architecture agar scalable untuk pengembangan ERP di masa depan tanpa overengineering.

3. Business Problem

3.1 Current Problems

Transaction Management

- Transaksi penjualan masih dicatat secara manual atau menggunakan POS sederhana.
- Riwayat transaksi sulit dilacak secara detail.

Inventory Problems

- Pengurangan stok sering tidak sinkron dengan transaksi penjualan.
- Tidak ada monitoring low stock secara real-time.
- Stock movement tidak terdokumentasi dengan baik.

Reporting Problems

- Laporan penjualan dibuat manual.
- Proses rekap membutuhkan waktu lama.
- Data antar divisi tidak sinkron.

User Access Problems

- Semua user memiliki akses yang sama.
- Tidak ada role-based access control yang proper.

Scalability Problems

- Sistem existing sulit berkembang ketika bisnis bertambah besar.
 - Struktur aplikasi tidak modular.
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4. Business Goals

4.1 Primary Goals

ELKash bertujuan untuk:

1. Mengintegrasikan proses bisnis utama dalam satu sistem.
 2. Mengurangi human error operasional.
 3. Menyediakan data transaksi dan inventory secara real-time.
 4. Meningkatkan efisiensi operasional bisnis.
 5. Menyediakan pelaporan bisnis yang cepat dan akurat.
 6. Membangun fondasi ERP modular yang scalable.
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4.2 Long-Term Goals

1. Mendukung multi-module ERP expansion.
 2. Mendukung purchasing workflow.
 3. Mendukung accounting integration.
 4. Menjadi enterprise-ready portfolio project.
 5. Mendukung future mobile integration melalui REST API architecture.
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5. Project Scope

5.1 V1 Scope

Included Modules

Auth Module

- Login
- Logout
- Session Management
- Role Based Access Control

POS Module

- Product Grid
- Cart
- Checkout
- Payment
- Invoice
- Transaction History

Inventory Module

- Product Management
- Stock Management
- Category Management
- Stock Movement

Dashboard Module

- Revenue Summary
- Sales Chart
- Low Stock Alert
- Best Seller Product
- Transaction Statistics

Reporting Module

- Daily Sales Report
- Monthly Sales Report
- Inventory Report
- Cashier Report
- PDF Export
- Excel Export

5.2 Future Scope

V2

- Purchasing
- Supplier
- Purchase Request
- Purchase Order
- Sales Order
- Approval Workflow

V3

- Accounting

- Journal
 - Ledger
 - Financial Statement
 - Profit/Loss
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6. Target Users

6.1 Super Admin

Responsibilities:

- Manage all system configurations
- Manage roles and permissions
- Monitor all operational activities

Needs:

- Full system access
 - High visibility dashboard
 - Audit and monitoring capability
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6.2 Admin

Responsibilities:

- Manage products
- Manage inventory
- Manage reports

Needs:

- Product management tools
 - Reporting access
 - Inventory monitoring
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6.3 Cashier

Responsibilities:

- Handle transactions
- Process customer payments
- Print receipts

Needs:

- Fast POS interface
 - Stable transaction flow
 - Simple and responsive UI
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6.4 Manager

Responsibilities:

- Monitor business performance
- Analyze reports
- Monitor sales and inventory

Needs:

- Dashboard analytics
 - Reporting tools
 - Operational insights
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7. Business Workflow

7.1 POS Transaction Flow

Customer creates order → Cashier inputs transaction → System validates cart → Checkout process → Payment confirmation → Save transaction → Reduce inventory stock → Generate invoice/receipt → Update dashboard & reporting

7.2 Inventory Workflow

Admin manages products → System updates stock data → Stock movement recorded → Low stock monitoring triggered → Inventory reflected on dashboard and reports

7.3 Reporting Workflow

Transaction data collected → System aggregates data → Generate operational reports → Export to PDF/Excel → Management review

8. Business Rules

Transaction Rules

- Transaction cannot proceed with empty cart.
- Product stock must be available before checkout.
- Completed transactions automatically reduce inventory stock.

Authentication Rules

- All users must login before accessing system.
- Access permissions based on RBAC roles.
- Session handled via Laravel Sanctum.

Inventory Rules

- Stock adjustments must create stock movement logs.
- Deleted products cannot remove historical transaction data.

Reporting Rules

- Reports generated from real-time transactional data.
 - Reports exportable to PDF and Excel.
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9. Operational Requirements

Availability

System should be accessible during business operational hours with minimal downtime.

Performance

POS transaction flow must remain responsive and fast during peak operational hours.

Security

- Authentication centralized via Laravel backend.
- Frontend must never directly access database.
- Protected API endpoints using Sanctum authentication.

Scalability

System architecture must support future ERP module expansion without major refactor.

10. Technology & Architecture Requirements

Frontend

- Nuxt 3
- Vue 3
- TypeScript
- Pinia
- TailwindCSS
- Shadcn Vue

Backend

- Laravel REST API
- Laravel Sanctum
- PostgreSQL (Supabase)
- Redis (Optional)

Deployment

Frontend

- Netlify
- Auto deploy via GitHub

Backend

- VPS Ubuntu + Nginx
 - SSL via Let's Encrypt
 - PM2/Supervisor
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11. Expected Business Impact

Operational Impact

- Faster transaction processing
- Better inventory accuracy
- Centralized business data

Financial Impact

- Reduced operational inefficiency
- Reduced stock discrepancy
- Faster reporting process

Management Impact

- Better decision making through dashboard analytics
- Improved operational visibility
- Better business monitoring

Technical Impact

- Scalable ERP foundation
 - Easier future feature expansion
 - Better maintainability through modular architecture
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12. Success Indicators

Business KPI

- Faster checkout process
- Reduced stock mismatch
- Faster report generation
- Reduced manual operational process

Technical KPI

- Stable REST API communication
 - Modular architecture implementation
 - Responsive frontend performance
 - Clean backend maintainability
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13. Constraints & Assumptions

Constraints

- Initial development focused on V1 only.
- Redis optional and only implemented when necessary.
- Accounting module excluded from initial release.

Assumptions

- Business uses stable internet connection.
 - Staff already familiar with basic POS operation.
 - Transactions processed in centralized system.
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14. Final Objective

ELKash harus menjadi:

- professional modular ERP,
- realistic business system,
- scalable architecture foundation,
- dan strong portfolio-grade enterprise application.

Fokus utama sistem adalah integrasi business process antar module, bukan sekadar banyak fitur.